EXPERIMENT - 8

Privilege Escalation Using TryHackMe

Step 1: Login to Platform

Open TryHackMe and log in

Step 2: Open Room

Search for **Linux Privilege Escalation** and join the room.

Step 3: Start AttackBox

Launch the AttackBox and open the terminal.

Step 4: Perform Enumeration

Check user details, system information, and permissions to identify possible vulnerabilities.

Step 5: Identify Vulnerabilities

Look for misconfigurations such as sudo permissions, SUID files, weak file access, or cron jobs.

Step 6: Escalate Privileges

Exploit the identified vulnerability to gain higher privileges (root access).

Step 7: Verify and Complete Tasks

Confirm root access and capture required flags to complete the exercise.

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Linux Privilege Escalation

Learn the fundamentals of Linux privilege escalation. From enumeration to exploitation, get hands-on with over 8 different privilege escalation techniques.

50 min | 161,943

Start AttackBox | Save Room | 5884 Recommend | Options

Room progress (30%)

- Task 1 Introduction
- Task 2 What is Privilege Escalation?
- Task 3 Enumeration
- Task 4 Automated Enumeration Tools
- Task 5 Privilege Escalation: Kernel Exploits
- Task 6 Privilege Escalation: Sudo

Below is a short example used to find files that have the SUID bit set. The SUID bit allows the file to run with the privilege level of the account that owns it, rather than the account which runs it. This allows for an interesting privilege escalation path,we will see in more details on task 6. The example below is given to complete the subject on the "find" command.

- `find / -perm -u=s -type f 2>/dev/null`: Find files with the SUID bit, which allows us to run the file with a higher privilege level than the current user.

General Linux Commands

As we are in the Linux realm, familiarity with Linux commands, in general, will be very useful. Please spend some time getting comfortable with commands such as `find`, `locate`, `grep`, `cut`, `sort`, etc.

Answer the questions below

What is the hostname of the target system?

wade7363 — Correct Answer

What is the Linux kernel version of the target system?

3.13.0-24-generic — Correct Answer

What Linux is this?

Ubuntu 14.04 LTS — Correct Answer

What version of the Python language is installed on the system?

2.7.6 — Correct Answer

What vulnerability seem to affect the kernel of the target system? (Enter a CVE number)

CVE-2015-1328 — Correct Answer

The Kernel exploit methodology is simple;

1. Identify the kernel version
2. Search and find an exploit code for the kernel version of the target system
3. Run the exploit

Although it looks simple, please remember that a failed kernel exploit can lead to a system crash. Make sure this potential outcome is acceptable within the scope of your penetration testing engagement before attempting a kernel exploit.

Research sources:

1. Based on your findings, you can use Google to search for an existing exploit code.
2. Sources such as https://www.cvedetails.com/ can also be useful.
3. Another alternative would be to use a script like LES (Linux Exploit Suggester) but remember that these tools can generate false positives (report a kernel vulnerability that does not affect the target system) or false negatives (not report any kernel vulnerabilities although the kernel is vulnerable).

Hints/Notes:

1. Being too specific about the kernel version when searching for exploits on Google, Exploit-db, or searchsploit
2. Be sure you understand how the exploit code works BEFORE you launch it. Some exploit codes can make changes on the operating system that would make them unsecured in further use or make irreversible changes to the system, creating problems later. Of course, these may not be great concerns within a lab or CTF environment, but these are absolute no-nos during a real penetration testing engagement.
3. Some exploits may require further interaction once they are run. Read all comments and instructions provided with the exploit code.
4. You can transfer the exploit code from your machine to the target system using the `SimpleHTTPServer` Python module and `wget` respectively.

Answer the questions below

find and use the appropriate kernel exploit to gain root privileges on the target system.

No answer needed ✓ Correct Answer

What is the content of the flag1.txt file?

THM-28392872729920 ✓ Correct Answer

Result

Privilege escalation was successfully performed, demonstrating how system misconfigurations can lead to unauthorized access and highlighting the importance of secure system practices.